Lecture 10: Types of Stereoscopic Displays

“Last class we learned how our brain perceives depth. But how do screens actually deliver two images to two eyes?”

Explain:

Different technologies use different tricks to separate left-eye and right-eye images.

What is a Stereoscopic Display?

Define simply:

A stereoscopic display presents two slightly different images separately to each eye to create depth illusion.

Anaglyph Displays (Red–Cyan Glasses)

How it works:

- Left image colored red
- Right image colored cyan
- Glasses filter colors so each eye sees only one image

Example:

- Old 3D movies
- Cheap 3D books

Pros:

- Very low cost
- Simple technology
- Works on any screen

Cons:

- Poor color quality
- Eye strain
- Not immersive

Tell students:

Good for demonstration, bad for real AR/VR.

Active Shutter Displays

How it works:

- Screen alternates left/right images rapidly
- Electronic glasses open/close each eye in sync

Explain simply:

Like blinking each eye 120 times per second.

Used in:

- High-end 3D TVs
- Professional simulators

Pros:

- Full color quality
- High resolution per eye

Cons:

- Glasses expensive
- Needs battery
- Can cause flicker discomfort

Passive Polarized Displays

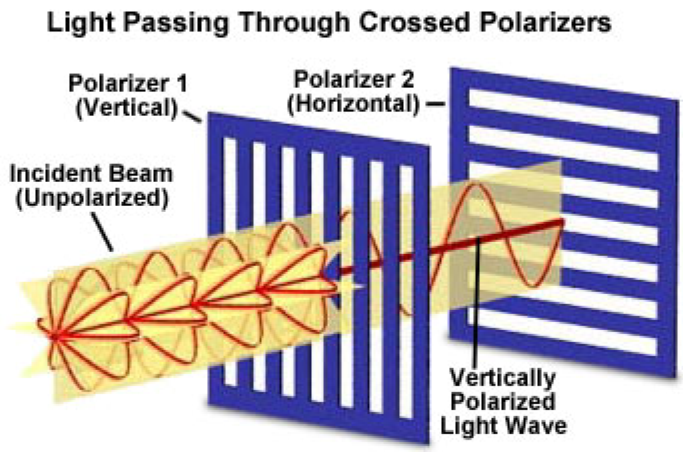

Figure 1

How it works:

- Uses polarization (light orientation)
- Glasses have different polarizing filters for each eye
- Screen sends two polarized images

Used in:

- IMAX 3D
- Cinemas
- Some VR caves (CAVE systems)

Pros:

- Lightweight glasses
- Cheaper than active
- Comfortable for long viewing

Cons:

- Lower resolution per eye
- Requires special screens

Head Mounted Displays (HMDs) – Core of VR

Explain:

HMDs are personal stereoscopic systems worn on the head.

How they work:

- Two screens (or one split screen)
- One image per eye
- Lenses enhance field of view
- Sensors track head movement

Examples:

- Meta Quest
- HTC Vive
- PlayStation VR
- Apple Vision Pro

Pros:

- Highly immersive
- Head tracking + stereoscopy
- Best for AR/VR

Cons:

- Expensive
- Can cause motion sickness
- Needs powerful GPU

📊 55–58 min → Comparison Table (Very exam-friendly)

Type	Glasses Needed	Cost	Quality	Immersion	Use Case
Anaglyph	Yes (red/cyan)	Very low	Poor	Low	Education demos
Active	Yes (electronic)	High	Very good	Medium	3D TVs
Passive	Yes (polarized)	Medium	Good	Medium	Cinemas
HMD	No external glasses	High	Excellent	Very high	AR/VR systems

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Recap + Quick Questions

Recap:

- All technologies try to separate left and right images
- Anaglyph = cheapest, worst quality
- Active/Passive = better but still limited
- HMDs = best solution for AR/VR

Quick oral questions:

- Why are anaglyph displays unsuitable for VR?
- Which technology gives highest immersion?
- Why do HMDs require powerful GPUs?